Phase 3 Derivation
Dynamic Organizational Choice, Bellman System, and Value-Function Existence

1 Dynamic Organizational Choice, Bellman System, and Value-Function Existence

This phase derives the fixed-price dynamic block that sits on top of the optimized static operating payoffs from Phase 2. Throughout this section we hold fixed a policy regime $M \in \{0, 1\}$ and fixed prices (w, p_m). Every existence, uniqueness, and policy statement below is therefore conditional on (w, p_m, M). This is a proof about the incumbent firms' Bellman system at fixed prices; it is not yet a proof of full stationary competitive equilibrium existence.

1.1 Benchmark Fidelity Check

Step 1: exact benchmark objects relevant for the present task. The authoritative sources for this phase are the timing block, notation table, and dynamic organizational-choice block in the benchmark manuscript, cross-checked against the benchmark PDF and the master version dictionary. The relevant benchmark objects are:

- $V_A(z; M)$, $V_B(z; M)$, and $V_C(z; M)$: endogenous dynamic value functions for incumbent firms. These are dynamic, endogenous, and structural-only objects.
- $\Pi_A(z; w, M)$, $\Pi_B(z; p_m, M)$, and $\Pi_C(z; p_m, w, M)$: optimized static operating payoffs imported from Phase 2. These are derived static objects, not primitives.
- $Q_A(\cdot \mid z, M)$, $Q_B(\cdot \mid z, M)$, and $Q_C(\cdot \mid z, M)$: primitive transition kernels for next period's capability state conditional on the current-period organizational form that the firm chooses.
- $\kappa_{sj}(M)$: primitive organizational switching cost from current type s to current-period type j.
- $W_{sj}(z; M)$: derived choice-specific value for a current type-s firm that chooses current-period organization j.
- $\mathcal{Z}_{sj}(M)$: derived benchmark choice region, defined as the set of states in which action j solves the Bellman problem for current type s.

Step 2: exact formulas, argument lists, and interpretation. The benchmark timing states that, within a period, an incumbent firm:

1. enters the period with current organizational type $s \in \{A, B, C\}$ and current capability state $z \in Z$;
2. chooses current-period organizational form $j \in \mathcal{J}(s)$;
3. conditional on that current-period organizational choice, chooses current operating controls and earns the corresponding optimized current operating payoff;
4. if it remains active, draws next period's capability state from the transition kernel associated with the chosen current-period organization.

The notation table fixes the operating-payoff argument lists exactly as

$$\Pi_A(z; w, M), \qquad \Pi_B(z; p_m, M), \qquad \Pi_C(z; p_m, w, M).$$

It fixes the value functions exactly as

$$V_A(z; M), \qquad V_B(z; M), \qquad V_C(z; M).$$

The interpretation is as follows. Π_A, Π_B, and Π_C summarize the best current operating payoff that each organizational form can attain at state z under fixed prices and regime M. The value functions summarize the total present discounted value of being an incumbent of each type before making the current-period organizational decision.

Step 3: can the present task be completed faithfully from the benchmark as written? Yes. At fixed prices we can:

1. construct the incumbent dynamic problem directly from the benchmark timing;
2. define the Bellman equations for V_A, V_B, and V_C without changing the benchmark objects;
3. place the Bellman system on an explicit complete metric space;
4. define a Bellman operator and prove that it is a contraction under the sup norm;
5. conclude existence and uniqueness by Banach's Fixed Point Theorem; and
6. describe the benchmark argmax correspondence, together with the extra convention required if one wants a single-valued measurable selector.

Step 4: inconsistency or ambiguity that must be flagged before proceeding. Two benchmark-fidelity issues must be stated explicitly.

1. **Argument-list tension.** The dynamic block in the manuscript sometimes writes the shorthand object $\Pi_j(z; w, p_m, M)$. However, the notation table and the Phase 2 canonical derivation retain the object-specific benchmark argument lists $\Pi_A(z; w, M)$, $\Pi_B(z; p_m, M)$, and $\Pi_C(z; p_m, w, M)$. Because benchmark fidelity forbids silently expanding argument lists, we do not create a new primitive object $\Pi_j(z; w, p_m, M)$. Instead we write the three active choice-specific values separately, one for A, one for B, and one for C.

2. **Tie-resolution tension.** The benchmark says that organizational choice is deterministic and that transition probabilities are generated by the states in which each action solves the Bellman problem. If ties occur, the corresponding solving regions overlap. Therefore a single-valued policy or a partition of the state space requires an extra tie-breaking convention or an extra uniqueness assumption. We do not silently impose either one inside the Bellman proof.

1.2 Part 1. Constructing the Dynamic Problem

1. Benchmark object restatement. At the beginning of the period, the incumbent firm has current organizational type $s \in \{A, B, C\}$ and current capability state $z \in Z$. For each current type s, the feasible current-period organizational choices are

$$\mathcal{J}(s) = \{A, B, C, \text{exit}\}.$$

For notational economy, when we write $V_s(z; M)$ with $s \in \{A, B, C\}$, we mean the corresponding component of the value-function triple:

$$V_s(z; M) = \begin{cases} V_A(z; M) & \text{if } s = A, \\ V_B(z; M) & \text{if } s = B, \\ V_C(z; M) & \text{if } s = C. \end{cases}$$

This is only shorthand; the benchmark primitives remain the three named objects V_A, V_B, and V_C.

2. Economic role. This block describes how an incumbent converts its current organizational identity and current capability into a current organizational decision. The decision is not only about current production or current research. It is about which organizational form the firm will operate under during the current period. That choice matters immediately, because current organization determines which current-period operating problem the firm is allowed to solve. It also matters dynamically, because current organization determines the law of motion for next period's capability state and therefore changes the continuation value.

3. Mathematical problem. Fix a current type $s \in \{A, B, C\}$, a current state $z \in Z$, fixed prices (w, p_m), and a fixed regime M. The incumbent chooses a current-period organization $j \in \mathcal{J}(s)$ to solve

$$V_s(z; M) = \max_{j \in \mathcal{J}(s)} W_{sj}(z; M).$$

The object on the left-hand side, $V_s(z; M)$, is the maximal attainable total value from starting the period as current type s at state z. The objects inside the maximum, $W_{sj}(z; M)$, are the values attached to each feasible current-period organizational action j.

4. Admissible set. For every current type s, the feasible set $\mathcal{J}(s)$ has exactly four elements:

$$\mathcal{J}(s) = \{A, B, C, \text{exit}\}.$$

Therefore the feasible set is:

- nonempty;
- finite;
- independent of z in the benchmark.

Because the set is finite, we do not need compact-choice-set machinery to guarantee that the maximum is attained. Once each choice-specific value $W_{sj}(z; M)$ is a finite real number, the maximum over the four feasible actions automatically exists.

5. Timing written slowly. The benchmark timing can be written as the following sequence.

1. The firm enters the period with inherited organization s and inherited state z.
2. The firm chooses its *current-period* organization j.
3. If $j \neq s$, the firm pays the one-time current-period switching cost $\kappa_{sj}(M)$.
4. Given the chosen current-period organization j, the firm solves the current operating problem associated with organization j and obtains optimized current payoff Π_j with the exact benchmark argument list appropriate for that j.
5. If $j \in \{A, B, C\}$, the firm remains active and next period's capability state z' is drawn from $Q_j(\cdot \mid z, M)$.
6. If $j =$ exit, the firm terminates operations and receives the benchmark exit value, which is normalized to zero.

6. Why the current-period organizational choice must come before the current-period operating controls. This point deserves a completely explicit explanation.

1. In the benchmark, the current-period operating payoff is not a generic function that is independent of organizational form. Instead, the current operating block depends on whether the firm operates as type A, type B, or type C during the current period.

2. A type-A firm solves the integrated-firm operating problem from Phase 2 and therefore earns $\Pi_A(z; w, M)$.

3. A type-B firm solves the research-specialized operating problem from Phase 2 and therefore earns $\Pi_B(z; p_m, M)$.

4. A type-C firm solves the manufacturing-specialized operating problem from Phase 2 and therefore earns $\Pi_C(z; p_m, w, M)$.

5. These three current operating payoffs are different functions because the benchmark assigns different same-period feasibility sets and different same-period commercialization possibilities to the three organizational forms.

Therefore the model must know the current-period organization before it can even identify which current-period operating problem is relevant. If the model reversed the timing and let the firm pick current operating controls before choosing current organization, then the phrase "current operating payoff" would be undefined: the model would not know whether to use Π_A, Π_B, or Π_C. That is why the current-period organizational choice comes first.

7. Choice-specific values defined one by one. We now write the benchmark choice-specific values slowly and separately, preserving the exact benchmark objects:

$$W_{sA}(z; M) = \Pi_A(z; w, M) - \kappa_{sA}(M) + \beta \int_Z V_A(z')\, Q_A(dz' \mid z, M), \tag{1}$$

$$W_{sB}(z; M) = \Pi_B(z; p_m, M) - \kappa_{sB}(M) + \beta \int_Z V_B(z')\, Q_B(dz' \mid z, M), \tag{2}$$

$$W_{sC}(z; M) = \Pi_C(z; p_m, w, M) - \kappa_{sC}(M) + \beta \int_Z V_C(z')\, Q_C(dz' \mid z, M), \tag{3}$$

$$W_{s,\text{exit}}(z; M) = 0. \tag{4}$$

8. Economic meaning of every term in $W_{sj}(z; M)$. Take the active choice $j = A$ as an example. The same interpretation applies to the other active choices.

1. The first term, $\Pi_A(z; w, M)$, is the best current-period operating payoff that the firm can obtain *during the current period* if it chooses to operate as type A.

2. The second term, $-\kappa_{sA}(M)$, is the current-period reorganization cost of moving from inherited organization s to current-period organization A. If $s = A$, this is the stay-as-A cost; in the benchmark we usually normalize $\kappa_{AA}(M) = 0$.

3. The third term, $\beta \int_Z V_A(z')Q_A(dz' \mid z, M)$, is the discounted expected continuation value. The integral means that the firm takes expectation with respect to the probability law $Q_A(\cdot \mid z, M)$ that governs next period's capability state when the current-period organizational choice is A.

For $j = B$ and $j = C$, the same logic applies with the appropriate type-specific operating payoff and type-specific transition kernel. For $j = \text{exit}$, the benchmark value is zero because the benchmark introduces neither a scrap value nor an exit subsidy nor an exit tax. The normalization is not forced by mathematics; it is a benchmark modeling choice and we preserve it exactly.

1.3 Part 2. Defining the Bellman Equations

1. Bellman equations derived from the generic dynamic problem. We now expand the generic dynamic problem type by type.

Current type A. Start from the generic expression

$$V_A(z; M) = \max_{j \in \mathcal{J}(A)} W_{Aj}(z; M).$$

Because $\mathcal{J}(A) = \{A, B, C, \text{exit}\}$, this becomes

$$V_A(z; M) = \max\big\{W_{AA}(z; M), W_{AB}(z; M), W_{AC}(z; M), W_{A,\text{exit}}(z; M)\big\}.$$

Now substitute the benchmark normalization $W_{A,\text{exit}}(z; M) = 0$:

$$V_A(z; M) = \max\big\{W_{AA}(z; M), W_{AB}(z; M), W_{AC}(z; M), 0\big\}. \tag{5}$$

Current type B. Again start from the generic expression

$$V_B(z; M) = \max_{j \in \mathcal{J}(B)} W_{Bj}(z; M).$$

Because $\mathcal{J}(B) = \{A, B, C, \text{exit}\}$, we obtain

$$V_B(z; M) = \max\big\{W_{BA}(z; M), W_{BB}(z; M), W_{BC}(z; M), W_{B,\text{exit}}(z; M)\big\}.$$

Substituting $W_{B,\text{exit}}(z; M) = 0$ yields

$$V_B(z; M) = \max\big\{W_{BA}(z; M), W_{BB}(z; M), W_{BC}(z; M), 0\big\}. \tag{6}$$

Current type C. Likewise,

$$V_C(z; M) = \max_{j \in \mathcal{J}(C)} W_{Cj}(z; M) = \max\big\{W_{CA}(z; M), W_{CB}(z; M), W_{CC}(z; M), W_{C,\text{exit}}(z; M)\big\}.$$

Substituting $W_{C,\text{exit}}(z; M) = 0$ gives

$$V_C(z; M) = \max\big\{W_{CA}(z; M), W_{CB}(z; M), W_{CC}(z; M), 0\big\}. \tag{7}$$

2. Why exit is normalized to zero. The exit option has to appear in the Bellman problem because the benchmark allows incumbent exit. However, the benchmark does not specify any scrap payoff, liquidation value, exit tax, or exit subsidy. Therefore the benchmark exit value is

$$W_{s,\text{exit}}(z; M) = 0 \qquad \text{for every current type } s \in \{A, B, C\}.$$

This means that all active organizational options are compared to a zero outside option. If an active option yields negative value while exit yields zero, exit becomes optimal.

3. What is recursive in the Bellman system. The Bellman equations are recursive because the unknown objects appear on both sides.

- On the left-hand side, $V_s(z; M)$ is the current value of beginning the period as current type s at state z.
- On the right-hand side, each active choice-specific value contains the continuation term $\int_Z V_j(z')Q_j(dz' \mid z, M)$.
- That continuation term depends on the very same value functions V_A, V_B, and V_C that we are trying to solve for.

So the Bellman system is not a collection of static algebraic equations. It is a system of functional equations. The unknowns are functions, not numbers. That is the reason we must next place the problem on a function space and define an operator whose fixed point is the value-function triple.

1.4 Part 3. Putting the Bellman System on a Function Space

1. Why a function space is needed. The Bellman system does not ask us to solve for three constants. It asks us to solve for three functions of z. To apply a fixed-point theorem, we need:

1. a space in which the candidate functions live;
2. a norm that measures the distance between two candidate triples;
3. an operator that takes one candidate triple and returns the next Bellman update.

Only after those objects are defined can we ask whether the operator maps the space into itself and whether repeated Bellman iteration converges.

2. Function space. Define

$$\mathcal{C}_b(Z) := \{f : Z \to \mathbb{R} \mid f \text{ is bounded and continuous}\}.$$

Because the benchmark state space Z is compact, continuity already implies boundedness. Nevertheless we keep the notation $\mathcal{C}_b(Z)$ because it highlights the two properties we want the value functions to have: continuity and boundedness.

The value triple will live in the product space

$$\mathcal{V} := \mathcal{C}_b(Z) \times \mathcal{C}_b(Z) \times \mathcal{C}_b(Z).$$

An arbitrary element of $\mathcal{V}$ is denoted by

$$U = (U_A, U_B, U_C),$$

where each component U_j is a bounded continuous candidate continuation value for organization j.

3. Norm. For a single bounded function $f : Z \to \mathbb{R}$, define the sup norm

$$\|f\|_\infty := \sup_{z \in Z} |f(z)|.$$

For a triple $U = (U_A, U_B, U_C) \in \mathcal{V}$, define the product-space norm

$$\|U\|_{\mathcal{V}} := \max\{\|U_A\|_\infty, \|U_B\|_\infty, \|U_C\|_\infty\}.$$

This norm says that two triples are close if and only if each component is uniformly close over the entire state space.

4. Auxiliary action-value indices used only to define the operator. The benchmark object $W_{sj}(z; M)$ is defined using the true value functions. To define a Bellman operator, we must first allow the continuation term to be evaluated at an arbitrary candidate triple $U \in \mathcal{V}$. For this purpose only, define the auxiliary functions

$$\mathcal{W}_{sA}(z; M; U) = \Pi_A(z; w, M) - \kappa_{sA}(M) + \beta \int_Z U_A(z')\, Q_A(dz' \mid z, M), \tag{8}$$

$$\mathcal{W}_{sB}(z; M; U) = \Pi_B(z; p_m, M) - \kappa_{sB}(M) + \beta \int_Z U_B(z')\, Q_B(dz' \mid z, M), \tag{9}$$

$$\mathcal{W}_{sC}(z; M; U) = \Pi_C(z; p_m, w, M) - \kappa_{sC}(M) + \beta \int_Z U_C(z')\, Q_C(dz' \mid z, M), \tag{10}$$

$$\mathcal{W}_{s,\text{exit}}(z; M; U) = 0. \tag{11}$$

This auxiliary notation does *not* replace the benchmark object. At the fixed point $V^* = (V_A^*, V_B^*, V_C^*)$, we recover the benchmark choice-specific values exactly:

$$W_{sj}(z; M) = \mathcal{W}_{sj}(z; M; V^*).$$

5. Bellman operator. Define the Bellman operator $\mathcal{T}_M : \mathcal{V} \to \mathcal{V}$ component by component:

$$(\mathcal{T}_M U)_A(z) = \max\big\{\mathcal{W}_{AA}(z; M; U), \mathcal{W}_{AB}(z; M; U), \mathcal{W}_{AC}(z; M; U), 0\big\}, \tag{12}$$

$$(\mathcal{T}_M U)_B(z) = \max\big\{\mathcal{W}_{BA}(z; M; U), \mathcal{W}_{BB}(z; M; U), \mathcal{W}_{BC}(z; M; U), 0\big\}, \tag{13}$$

$$(\mathcal{T}_M U)_C(z) = \max\big\{\mathcal{W}_{CA}(z; M; U), \mathcal{W}_{CB}(z; M; U), \mathcal{W}_{CC}(z; M; U), 0\big\}. \tag{14}$$

6. Why a fixed point of $\mathcal{T}_M$ is exactly a solution to the Bellman system. Suppose first that $U \in \mathcal{V}$ is a fixed point of $\mathcal{T}_M$. Then by definition,

$$U = \mathcal{T}_M U.$$

This means that, for every $z \in Z$,

$$U_A(z) = (\mathcal{T}_M U)_A(z), \qquad U_B(z) = (\mathcal{T}_M U)_B(z), \qquad U_C(z) = (\mathcal{T}_M U)_C(z).$$

Substituting the definition of $\mathcal{T}_M$ shows that U satisfies the three Bellman equations with U on both sides. Conversely, if a triple U satisfies the three Bellman equations, then each component of U equals the corresponding component of $\mathcal{T}_M U$, so $U = \mathcal{T}_M U$. Therefore:

$$\text{solving the Bellman system} \quad \Longleftrightarrow \quad \text{finding a fixed point of } \mathcal{T}_M.$$

1.5 Part 4. Proving Well-Posedness of the Fixed-Price Bellman System

1. Assumptions used. Benchmark assumptions or objects carried directly from the benchmark and Phase 2:

(B1) The capability state space $Z \subset \mathbb{R}_+$ is compact.

(B2) For fixed (w, p_m, M), Phase 2 has already delivered bounded and continuous operating payoffs:

$$\Pi_A(\cdot; w, M), \qquad \Pi_B(\cdot; p_m, M), \qquad \Pi_C(\cdot; p_m, w, M) \in \mathcal{C}_b(Z).$$

(B3) For each organizational form $j \in \{A, B, C\}$, $Q_j(\cdot \mid z, M)$ is a probability kernel on $(Z, \mathcal{B}(Z))$.

(B4) The discount factor satisfies $\beta \in (0, 1)$.

Supplemental regularity assumptions used only for the Bellman proof:

(R1) The switching costs are finite real numbers: $|\kappa_{sj}(M)| < \infty$ for all $s, j \in \{A, B, C\}$.

(R2) Each transition kernel is Feller at fixed (M, j): for every $f \in \mathcal{C}_b(Z)$, the map

$$z \mapsto \int_Z f(z')\, Q_j(dz' \mid z, M)$$

is continuous on Z.

The benchmark sources define the kernels Q_j but do not themselves spell out the Feller property. We therefore state R2 explicitly as a supplemental regularity assumption rather than silently importing it.

2. A theorem we will use inside the completeness proof. We use the **Uniform Limit Theorem**: if X is a metric space, each $f_n : X \to \mathbb{R}$ is continuous, and $f_n \to f$ uniformly on X, then the limit function f is continuous on X. The assumptions are therefore:

1. the domain is a metric space;
2. every approximating function is continuous;
3. convergence is uniform rather than merely pointwise.

Lemma 1 (Completeness of the product space)**.** *Under B1, the normed space* $(\mathcal{V}, \|\cdot\|_{\mathcal{V}})$ *is complete.*

Proof. Take any Cauchy sequence $\{U^n\}_{n\geq 1}$ in $(\mathcal{V}, \|\cdot\|_{\mathcal{V}})$, where

$$U^n = (U_A^n, U_B^n, U_C^n).$$

Because the product norm is the maximum of the three component sup norms, the Cauchy property means:

$$\forall \varepsilon > 0,\ \exists N(\varepsilon) \text{ such that for all } n, m \geq N(\varepsilon), \quad \|U^n - U^m\|_{\mathcal{V}} < \varepsilon.$$

By definition of $\|\cdot\|_{\mathcal{V}}$, this implies

$$\|U_A^n - U_A^m\|_\infty < \varepsilon, \qquad \|U_B^n - U_B^m\|_\infty < \varepsilon, \qquad \|U_C^n - U_C^m\|_\infty < \varepsilon$$

whenever $n, m \geq N(\varepsilon)$. So each component sequence is Cauchy in the sup norm.

Fix one component, say $\{U_A^n\}$. For each fixed state $z \in Z$, the sequence of real numbers $\{U_A^n(z)\}_{n\geq 1}$ is Cauchy in $\mathbb{R}$, because

$$|U_A^n(z) - U_A^m(z)| \leq \|U_A^n - U_A^m\|_\infty.$$

Since $\mathbb{R}$ is complete, for each $z \in Z$ there exists a real number $U_A(z)$ such that

$$U_A^n(z) \to U_A(z) \qquad \text{as } n \to \infty.$$

This defines a candidate limit function $U_A : Z \to \mathbb{R}$ pointwise.

We now show that the convergence is actually uniform. Fix $\varepsilon > 0$ and choose $N(\varepsilon)$ as above. For any $n \geq N(\varepsilon)$ and any $m \geq N(\varepsilon)$,

$$\|U_A^n - U_A^m\|_\infty < \varepsilon.$$

Hold n fixed and let $m \to \infty$. For each $z \in Z$,

$$|U_A^n(z) - U_A(z)| = \lim_{m\to\infty} |U_A^n(z) - U_A^m(z)| \leq \varepsilon.$$

Because this bound holds for every $z \in Z$, we obtain

$$\|U_A^n - U_A\|_\infty \leq \varepsilon \qquad \text{for all } n \geq N(\varepsilon).$$

Thus $U_A^n \to U_A$ uniformly.

Next we show that U_A is bounded and continuous. Each U_A^n belongs to $\mathcal{C}_b(Z)$, so it is bounded and continuous. Uniform convergence plus the Uniform Limit Theorem implies that U_A is continuous. Boundedness follows because U_A^N is bounded for any fixed N, and

$$|U_A(z)| \leq |U_A(z) - U_A^N(z)| + |U_A^N(z)| \leq \|U_A - U_A^N\|_\infty + \|U_A^N\|_\infty$$

for every $z \in Z$. The right-hand side is finite, so U_A is bounded.

Exactly the same argument constructs bounded continuous limits U_B and U_C for the other two component sequences. Define

$$U := (U_A, U_B, U_C).$$

Then $U \in \mathcal{V}$. Moreover, because each component converges uniformly,

$$\|U^n - U\|_{\mathcal{V}} = \max\{\|U_A^n - U_A\|_\infty, \|U_B^n - U_B\|_\infty, \|U_C^n - U_C\|_\infty\} \longrightarrow 0.$$

So every Cauchy sequence in $\mathcal{V}$ converges to an element of $\mathcal{V}$. Therefore $(\mathcal{V}, \|\cdot\|_{\mathcal{V}})$ is complete. □

Lemma 2 (The Bellman operator maps $\mathcal{V}$ into itself)**.** *Under assumptions B1–B4 and R1–R2, for every $U \in \mathcal{V}$ we have $\mathcal{T}_M U \in \mathcal{V}$.*

Proof. Fix an arbitrary $U = (U_A, U_B, U_C) \in \mathcal{V}$. We must show that each component of $\mathcal{T}_M U$ is both bounded and continuous on Z.

Step 1: each continuation term is well defined and bounded. Take an active organizational form $j \in \{A, B, C\}$. Because $U_j \in \mathcal{C}_b(Z)$, the sup norm $\|U_j\|_\infty$ is finite. For any current state $z \in Z$,

$$\begin{aligned}
\left|\int_Z U_j(z')\, Q_j(dz' \mid z, M)\right| &\leq \int_Z |U_j(z')|\, Q_j(dz' \mid z, M) && \text{by the triangle inequality for integrals} \\
&\leq \int_Z \|U_j\|_\infty\, Q_j(dz' \mid z, M) && \text{because } |U_j(z')| \leq \|U_j\|_\infty \\
&= \|U_j\|_\infty \int_Z Q_j(dz' \mid z, M) \\
&= \|U_j\|_\infty,
\end{aligned}$$

where the last equality uses the fact that $Q_j(\cdot \mid z, M)$ is a probability measure. So each continuation term is a finite real number.

Step 2: each active choice-specific index is bounded. Define two finite constants:

$$\overline{\Pi} := \max\Big\{\sup_{z\in Z} |\Pi_A(z; w, M)|,\ \sup_{z\in Z} |\Pi_B(z; p_m, M)|,\ \sup_{z\in Z} |\Pi_C(z; p_m, w, M)|\Big\},$$

and

$$\overline{\kappa} := \max_{s,j\in\{A,B,C\}} |\kappa_{sj}(M)|.$$

These constants are finite by B2 and R1. Now consider, for example, the active index $\mathcal{W}_{sA}(z; M; U)$:

$$\begin{aligned}
|\mathcal{W}_{sA}(z; M; U)| &= \left|\Pi_A(z; w, M) - \kappa_{sA}(M) + \beta \int_Z U_A(z')\, Q_A(dz' \mid z, M)\right| \\
&\leq |\Pi_A(z; w, M)| + |\kappa_{sA}(M)| + \beta \left|\int_Z U_A(z')\, Q_A(dz' \mid z, M)\right| \\
&\leq \overline{\Pi} + \overline{\kappa} + \beta \|U_A\|_\infty \\
&\leq \overline{\Pi} + \overline{\kappa} + \beta \|U\|_{\mathcal{V}}.
\end{aligned}$$

The same calculation applies to $\mathcal{W}_{sB}$ and $\mathcal{W}_{sC}$. For the exit action,

$$|\mathcal{W}_{s,\text{exit}}(z; M; U)| = 0.$$

Therefore every term inside the maximum that defines $(\mathcal{T}_M U)_s(z)$ is bounded uniformly over z. **Step 3: each component of $\mathcal{T}_M U$ is bounded.** Fix $s \in \{A, B, C\}$. For each $z \in Z$, $(\mathcal{T}_M U)_s(z)$ is the maximum of four real numbers. Hence

$$\begin{aligned}
|(\mathcal{T}_M U)_s(z)| &= \left|\max\{\mathcal{W}_{sA}(z; M; U), \mathcal{W}_{sB}(z; M; U), \mathcal{W}_{sC}(z; M; U), 0\}\right| \\
&\leq \max\Big\{|\mathcal{W}_{sA}(z; M; U)|,\, |\mathcal{W}_{sB}(z; M; U)|,\, |\mathcal{W}_{sC}(z; M; U)|,\, 0\Big\} \\
&\leq \overline{\Pi} + \overline{\kappa} + \beta \|U\|_{\mathcal{V}}.
\end{aligned}$$

So $(\mathcal{T}_M U)_s$ is bounded.

Step 4: each active choice-specific index is continuous. Fix an active organizational form $j \in \{A, B, C\}$. By R2, the map

$$z \mapsto \int_Z U_j(z')\, Q_j(dz' \mid z, M)$$

is continuous on Z because $U_j \in \mathcal{C}_b(Z)$. By B2, the map $z \mapsto \Pi_j(\cdot)$ with the correct benchmark argument list is continuous. The switching cost $\kappa_{sj}(M)$ is a constant with respect to z. The sum of continuous functions is continuous. Therefore each active index

$$z \mapsto \mathcal{W}_{sj}(z; M; U)$$

is continuous on Z. The exit index is the constant zero function, so it is also continuous.

Step 5: the maximum of the four continuous indices is continuous. Define the four-dimensional map

$$m(a_1, a_2, a_3, a_4) := \max\{a_1, a_2, a_3, a_4\}.$$

This map is continuous on $\mathbb{R}^4$. Since each of the four indices entering $(\mathcal{T}_M U)_s(z)$ is continuous in z, the composition of those four continuous indices with the continuous map m is continuous. Therefore $(\mathcal{T}_M U)_s$ is continuous.

We have shown that each component $(\mathcal{T}_M U)_s$ is bounded and continuous. Hence $(\mathcal{T}_M U)_s \in \mathcal{C}_b(Z)$ for $s = A, B, C$. Therefore $\mathcal{T}_M U \in \mathcal{V}$. □

3. An inequality used in the contraction proof. We will use the elementary inequality

$$\left|\max_i x_i - \max_i y_i\right| \le \max_i |x_i - y_i|$$

for two finite vectors $(x_i)_i$ and $(y_i)_i$. Because this inequality is often quoted too quickly, we prove it explicitly.

Let i^* be an index such that $x_{i^*} = \max_i x_i$. Then

$$\begin{aligned}\max_i x_i - \max_i y_i &= x_{i^*} - \max_i y_i \\ &\le x_{i^*} - y_{i^*} \\ &\le |x_{i^*} - y_{i^*}| \\ &\le \max_i |x_i - y_i|.\end{aligned}$$

If we interchange the roles of x and y, we also get

$$\max_i y_i - \max_i x_i \le \max_i |x_i - y_i|.$$

Combining the two displayed inequalities yields

$$\left|\max_i x_i - \max_i y_i\right| \le \max_i |x_i - y_i|.$$

Lemma 3 (Contraction under the sup norm)**.** *Under assumptions B1–B4 and R1–R2, for all $U, \widehat{U} \in \mathcal{V}$,*

$$\|\mathcal{T}_M U - \mathcal{T}_M \widehat{U}\|_{\mathcal{V}} \le \beta \|U - \widehat{U}\|_{\mathcal{V}}.$$

Proof. Fix an arbitrary current type $s \in \{A, B, C\}$ and an arbitrary state $z \in Z$. Write the four action values entering $(\mathcal{T}_M U)_s(z)$ as a finite vector, and the four action values entering $(\mathcal{T}_M \widehat{U})_s(z)$ as another finite vector. By the inequality proved above,

$$\left|(\mathcal{T}_M U)_s(z) - (\mathcal{T}_M \widehat{U})_s(z)\right| \le \max_{j \in \{A,B,C,\text{exit}\}} \left|\mathcal{W}_{sj}(z; M; U) - \mathcal{W}_{sj}(z; M; \widehat{U})\right|.$$

So it remains to bound the difference for each action j.

Exit action. For $j = \text{exit}$, both auxiliary values equal zero by definition. Hence

$$\left|\mathcal{W}_{s,\text{exit}}(z; M; U) - \mathcal{W}_{s,\text{exit}}(z; M; \widehat{U})\right| = 0.$$

Active actions. Now fix an active action $j \in \{A, B, C\}$. The current operating payoff and the switching cost cancel because they do not depend on the candidate continuation triple. Therefore

$$\begin{aligned}\mathcal{W}_{sj}(z; M; U) - \mathcal{W}_{sj}(z; M; \widehat{U}) &= \beta \int_Z U_j(z')\, Q_j(dz' \mid z, M) - \beta \int_Z \widehat{U}_j(z')\, Q_j(dz' \mid z, M) \\ &= \beta \int_Z \big(U_j(z') - \widehat{U}_j(z')\big)\, Q_j(dz' \mid z, M).\end{aligned}$$

Take absolute values:

$$\begin{aligned}
\left|\mathcal{W}_{sj}(z;M;U) - \mathcal{W}_{sj}(z;M;\widehat{U})\right| &= \beta \left| \int_Z \big(U_j(z') - \widehat{U}_j(z')\big)\, Q_j(dz' \mid z, M) \right| \\
&\le \beta \int_Z \left| U_j(z') - \widehat{U}_j(z') \right| Q_j(dz' \mid z, M) \\
&\le \beta \int_Z \|U_j - \widehat{U}_j\|_\infty \, Q_j(dz' \mid z, M) \\
&= \beta \|U_j - \widehat{U}_j\|_\infty \int_Z Q_j(dz' \mid z, M) \\
&= \beta \|U_j - \widehat{U}_j\|_\infty \\
&\le \beta \|U - \widehat{U}\|_{\mathcal{V}}.
\end{aligned}$$

Because this bound is valid for every active action and the exit difference is zero, we obtain

$$\left|(\mathcal{T}_M U)_s(z) - (\mathcal{T}_M \widehat{U})_s(z)\right| \le \beta \|U - \widehat{U}\|_{\mathcal{V}}.$$

The right-hand side does not depend on z or on s. Therefore we can take the supremum over $z \in Z$ and then the maximum over $s \in \{A, B, C\}$ to conclude that

$$\|\mathcal{T}_M U - \mathcal{T}_M \widehat{U}\|_{\mathcal{V}} \le \beta \|U - \widehat{U}\|_{\mathcal{V}}.$$

This is exactly the contraction inequality. □

4. The exact fixed-point theorem used and its assumptions. We now state the theorem explicitly rather than referring vaguely to "standard arguments."

Theorem 1 (Banach's Fixed Point Theorem)**.** *Let (X, d) be a complete metric space. Let $T : X \to X$ be a self-map. Suppose there exists a constant $q \in [0, 1)$ such that*

$$d(Tx, Ty) \le q\, d(x, y) \qquad \text{for all } x, y \in X.$$

Then:

1. *there exists a unique fixed point $x^* \in X$ such that $Tx^* = x^*$;*
2. *for every starting point $x_0 \in X$, the iterates $T^n x_0$ converge to x^*.*

The assumptions of this theorem match our Bellman problem as follows:

1. the complete metric space is $(\mathcal{V}, \|\cdot\|_{\mathcal{V}})$ by the completeness lemma;
2. the self-map is $\mathcal{T}_M : \mathcal{V} \to \mathcal{V}$ by the self-map lemma;
3. the contraction constant is $q = \beta$, which lies in $[0, 1)$ by B4 and by the contraction lemma.

Theorem 2 (Existence and uniqueness of the fixed-price value functions)**.** *Under assumptions B1–B4 and R1–R2, there exists a unique triple*

$$V^* = (V_A^*, V_B^*, V_C^*) \in \mathcal{V}$$

such that

$$\mathcal{T}_M V^* = V^*.$$

Equivalently, the benchmark Bellman system (5)–(7) *has a unique solution in* $\mathcal{V}$*.*

Proof. By the first lemma, $(\mathcal{V}, \|\cdot\|_{\mathcal{V}})$ is complete. By the second lemma, $\mathcal{T}_M$ maps $\mathcal{V}$ into itself. By the third lemma, $\mathcal{T}_M$ is a contraction with modulus $\beta < 1$. All assumptions of Banach's Fixed Point Theorem are therefore satisfied. The theorem gives a unique fixed point $V^* \in \mathcal{V}$ and convergence of value iteration from any initial guess. Since fixed points of $\mathcal{T}_M$ are exactly solutions to the Bellman system, the Bellman system has a unique solution in $\mathcal{V}$. □

1.6 Part 5. Policy Existence, Set-Valued Versus Single-Valued Policies, and Measurable Selection

1. Benchmark policy objects. Let V^* denote the unique fixed-price value-function triple. Using V^*, define the benchmark argmax correspondence for each current type s by

$$\Gamma_s(z; M) := \arg\max_{j \in \{A,B,C,\text{exit}\}} \mathcal{W}_{sj}(z; M; V^*).$$

This correspondence tells us which current-period organizational actions solve the Bellman problem at state z for a current type-s firm.

The benchmark choice-region object is then

$$\mathcal{Z}_{sj}(M) := \{z \in Z : j \in \Gamma_s(z; M)\}.$$

This is exactly the benchmark definition. We are not replacing it with a selector-based alternative.

Proposition 1 (Policy existence and measurability)**.** *Fix* (w, p_m, M) *and suppose assumptions B1–B4 and R1–R2 hold.*

1. *For every current type* $s \in \{A, B, C\}$ *and every state* $z \in Z$*, the set* $\Gamma_s(z; M)$ *is nonempty.*

2. *For every current type* s *and action* $j \in \{A, B, C, \text{exit}\}$*, the benchmark choice region* $\mathcal{Z}_{sj}(M)$ *is a Borel subset of* Z*.*

3. *If exactly one action attains the maximum at state* z*, then the policy is single-valued at that state.*

4. *If more than one action attains the maximum at state* z*, then the benchmark policy object is set-valued at that state.*

5. *If one adds an explicit deterministic tie-breaking order, then one can construct a single-valued measurable selector. That selector is an auxiliary implementation device, not the benchmark primitive object.*

Proof. **Step 1: argmax existence.** Fix a current type s and a state z. The Bellman problem compares the four real numbers

$$\mathcal{W}_{sA}(z; M; V^*), \qquad \mathcal{W}_{sB}(z; M; V^*), \qquad \mathcal{W}_{sC}(z; M; V^*), \qquad 0.$$

These are finite real numbers because $V^* \in \mathcal{V}$ and the choice-specific values are therefore well defined. A finite nonempty set of real numbers always attains its maximum. Hence $\Gamma_s(z; M)$ is nonempty. **Step 2: an explicit formula for the benchmark choice regions.** Fix s and j. By definition, $z \in \mathcal{Z}_{sj}(M)$ if and only if action j attains at least as much value as every competitor $\ell \in \{A, B, C, \text{exit}\}$. Therefore

$$\mathcal{Z}_{sj}(M) = \bigcap_{\ell \in \{A,B,C,\text{exit}\}} \{z \in Z : \mathcal{W}_{sj}(z; M; V^*) \geq \mathcal{W}_{s\ell}(z; M; V^*)\}.$$

This formula is useful because it rewrites the choice region as a finite intersection of comparison sets.

Step 3: measurability of the benchmark choice regions. Fix a competitor ℓ. Consider the difference function

$$D_{sj\ell}(z) := \mathcal{W}_{sj}(z; M; V^*) - \mathcal{W}_{s\ell}(z; M; V^*).$$

Each $\mathcal{W}_{sj}(\cdot; M; V^*)$ is continuous by the same reasoning used in the self-map proof. The difference of two continuous functions is continuous. Hence $D_{sj\ell}$ is continuous. The comparison set

$$\{z \in Z : \mathcal{W}_{sj}(z; M; V^*) \geq \mathcal{W}_{s\ell}(z; M; V^*)\} = D_{sj\ell}^{-1}\big([0, \infty)\big)$$

is therefore closed, because $[0, \infty)$ is closed in $\mathbb{R}$ and the preimage of a closed set under a continuous function is closed. A closed subset of the compact metric space Z is Borel. Since $\mathcal{Z}_{sj}(M)$ is a finite intersection of such Borel sets, $\mathcal{Z}_{sj}(M)$ is Borel measurable.

Step 4: single-valued versus set-valued policy. If exactly one action j attains the maximum, then the argmax set contains exactly one element:

$$\Gamma_s(z; M) = \{j\}.$$

In that case the policy is single-valued at state z. If two or more actions attain the same maximal value, then $\Gamma_s(z; M)$ contains multiple elements, so the benchmark policy object is set-valued at that state.

Step 5: an explicit measurable selector after adding a tie-breaking convention. Suppose we add the deterministic priority order

$$A \succ B \succ C \succ \text{exit}.$$

We can then define four measurable sets:

$$\begin{aligned}
G_{sA}(M) &:= \mathcal{Z}_{sA}(M), \\
G_{sB}(M) &:= \mathcal{Z}_{sB}(M) \cap \mathcal{Z}_{sA}(M)^c, \\
G_{sC}(M) &:= \mathcal{Z}_{sC}(M) \cap \mathcal{Z}_{sA}(M)^c \cap \mathcal{Z}_{sB}(M)^c, \\
G_{s,\text{exit}}(M) &:= \mathcal{Z}_{s,\text{exit}}(M) \cap \mathcal{Z}_{sA}(M)^c \cap \mathcal{Z}_{sB}(M)^c \cap \mathcal{Z}_{sC}(M)^c.
\end{aligned}$$

These sets are Borel because they are constructed from the Borel sets $\mathcal{Z}_{sj}(M)$ using finite intersections and complements.

Now define the tie-broken policy function

$$g_s(z; M) = \begin{cases} A & \text{if } z \in G_{sA}(M), \\ B & \text{if } z \in G_{sB}(M), \\ C & \text{if } z \in G_{sC}(M), \\ \text{exit} & \text{if } z \in G_{s,\text{exit}}(M). \end{cases}$$

Because the original argmax set is nonempty at every state, every state belongs to at least one of the original choice regions $\mathcal{Z}_{sj}(M)$. The sequential set subtraction assigns the highest-priority maximizing action at each state. Hence

$$g_s(z; M) \in \Gamma_s(z; M) \qquad \text{for every } z \in Z.$$

The function g_s is measurable because it is finite-valued and each inverse image

$$g_s^{-1}(\{A\}) = G_{sA}(M), \quad g_s^{-1}(\{B\}) = G_{sB}(M), \quad g_s^{-1}(\{C\}) = G_{sC}(M), \quad g_s^{-1}(\{\text{exit}\}) = G_{s,\text{exit}}(M)$$

is Borel measurable. □

2. Which theorem did we use for measurable selection? None. Because the action set is finite, we do not need to invoke Kuratowski–Ryll-Nardzewski or Berge's Maximum Theorem to obtain a measurable selector. We constructed the selector directly from measurable benchmark choice regions after adding an explicit tie-breaking convention. This keeps the logical dependence completely transparent.

3. Does the benchmark itself assume deterministic policies? Yes, in the limited sense that the benchmark does not add random utility shocks, logit smoothing, or mixed strategies. However, before ties are resolved, the benchmark policy object is the argmax correspondence Γ_s or, equivalently, the family of benchmark choice regions $\mathcal{Z}_{sj}(M)$. So the correct benchmark statement is:

- the model is non-random and deterministic once a maximizing action has been selected;
- ties must still be handled explicitly if one wants a single-valued policy function or disjoint transition regions.

4. Important fidelity note. The selector g_s constructed above is *not* the benchmark object. It is an auxiliary object that becomes available only after we impose an extra tie-breaking convention. Therefore we do not silently replace benchmark choice regions by tie-broken selector regions. If a later phase requires a row-stochastic transition matrix or a single-valued organizational policy function, that later phase must say explicitly whether it assumes uniqueness almost everywhere or adopts a particular tie-breaking rule.

1.7 What did we just do, and why does it matter?

We started from the benchmark's economic statement of the incumbent problem and converted it into a mathematical object that can actually be analyzed. The raw Bellman equations are recursive equations in unknown functions. Writing them down is only the first step. To know that the model is mathematically well posed, we must know that those equations have a solution and that the solution is unique. That is why we introduced the function space $\mathcal{V}$, the norm $\|\cdot\|_{\mathcal{V}}$, and the Bellman operator $\mathcal{T}_M$.

Why did we need a Bellman operator rather than just the Bellman equations themselves? Because Banach's Fixed Point Theorem applies to maps from a metric space into itself. The Bellman equations are a statement of equilibrium between today's value and tomorrow's continuation value, but the Bellman operator is the device that turns that verbal recursion into a self-map on a space of candidate value functions. Once we define the self-map, we can ask precise questions: does it stay inside the chosen function space, and does it shrink distances?

Why do we care about contraction? Contraction is the property that makes the recursion stable. Suppose two researchers start from two different guesses for the value functions. If one Bellman update shrinks the distance between those guesses by at least the factor β, and if $\beta < 1$, then repeated Bellman updates force the two sequences of guesses to move toward one another. This means that the recursive problem does not wander among multiple incompatible candidate values. Instead it pulls every initial guess toward a single fixed point. That is the mathematical reason value iteration works in this block.

Why does uniqueness matter for the economics that follow? Every later dynamic or equilibrium object is built on top of the value functions. Entry values depend on the value functions. Organizational choice regions depend on comparing choice-specific values, which themselves contain continuation values. Stationary distributions depend on induced policy behavior and transition probabilities. Market-clearing prices in the stationary equilibrium block will eventually have to be consistent with those induced objects. If the Bellman system had multiple fixed points, then later objects could inherit that multiplicity. Uniqueness eliminates that ambiguity and makes the later phases of the model logically determinate conditional on fixed prices.

The policy discussion matters for a separate reason. The fixed-point theorem gives us a unique value-function triple, but it does not automatically give us a unique action at every state. If ties occur, then the benchmark policy object is set-valued even though the value functions are unique. That distinction is important. The value functions can be perfectly well defined even when the induced policy correspondence is multi-valued. If later equilibrium construction needs a single-valued policy, tie handling must be added openly rather than hidden inside the notation.

1.8 End-of-Section Recap

- What has been established mathematically: holding (w, p_m, M) fixed, the benchmark Bellman system for (V_A, V_B, V_C) can be written as a self-map $\mathcal{T}_M$ on the complete metric space $\mathcal{V} = \mathcal{C}_b(Z)^3$; the operator preserves boundedness and continuity and satisfies the contraction inequality with modulus β; therefore the Bellman system has a unique fixed-price solution in $\mathcal{V}$.

- What assumptions were used: compact state space, Phase 2 continuity and boundedness of the optimized operating payoffs, probability-kernel structure for Q_A, Q_B, Q_C, discount factor $\beta \in (0, 1)$, finite switching costs, and the supplemental Feller continuity of the transition kernels.

- What remains to be derived next: later phases must use these value functions to construct entry values, stationary measures, market-clearing conditions, and the stationary equilibrium definition, while handling ties explicitly whenever a single-valued organizational policy or a row-stochastic transition matrix is needed.